MILITARY SPECIFICATION

BORING MACHINES, JIG VERTICAL, DOUBLE COLUMN, BRIDGE TYPE, FIXED OR TRAVELING CROSS-RAIL NUMERICALLY CONTROLLED (NC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers numerically controlled vertical jig boring machines, double column, bridge type having a traveling cross-rail with equipment and accessories specified herein.

1.2 Classification. This specification covers bridge type jig boring machines of the following sizes based on spindle cross travel and table travel. The sizes supplied shall be as specified (see 6.2.1)

Size 1 - 18 x 25-inch spindle cross travel and table travel
Size 2 - 23 x 35-inch spindle cross travel and table travel
Size 3 - 28 x 40-inch spindle cross travel and table travel
Size 4 - 40 x 60-inch spindle cross travel and table travel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3411

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA) (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified,

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the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care and Use.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC P-1 - Pneumatic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpart Drive, McLean, VA 22101.)

ELECTRONICS INDUSTRIES ASSOCIATION (EIA) STANDARD

RS-227 - One Inch Perforated Paper Tape.

RS-232 - Interface Between Data Terminal Equipment and Data Communications Equipment Employing Serial Binavy Data Interchange.

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RS-244 - Character Codes for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motor Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Perforated Tape Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Construction Standards - Numerical Machine Tool Control.

RS-358 - Subset of USA Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (IS0)

ISO 54 - Cylindrical Gears for General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or though libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 end 6.3).

3.2 Design. The machine shall be the manufacturer's current design for performing precision machining operations directed by a numerical control unit, The machine shall be capable of precision boring, drilling, and tapping operations and milling. Tools shall be manually inserted and removed from the spindle. Speeds, feeds, start/stop, clamp/unclamp, work table, quill, and horizontal spindle carriage travel shall be operator controlled from a console or pendant and by automatic control. The machine shall be of double column construction having a traveling cross-rail. The traveling cross-rail shall have vertical ways mounted on the columns to provide traverse of the cross-rail. Cross-rails shall have ways for traverse of the spindle head. Traverse of the spindle shall be across the longitudinal motion of the table. The table and operating mechanism thereof shall be mounted on the base of the machine. The design shall include all components, parts and features necessary to meet the performance requirements specified herein.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI systems of measurement. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanu-

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factured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The jig boring machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA CFR 1910 and ANSI B11.8 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be designed and constructed that, under the operating. service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation.

3.2.9 Lubrication. An automatic lubricating system shall be provided for l ll parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by oil level sight gage, a low oil level warning light or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The jig boring machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

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3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No processing such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All threaded fasteners shall comply with FED-STD-H28. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the jig boring machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the jig boring machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.0001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials and dial diameters shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem class and the emissions susceptibility requirements shall be as specified.

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3.4 Components. The jig boring machine shall consist primarily of a base, columns, fixed or traveling cross-rail, spindle carrier, spindle and quill assembly, worktable, saddle driving mechanisms, and operating controls.

3.4.1 Base. The base shall be a casting or weldment ribbed and braced to minimize distortion, deflection, and vibration. The base shall form a foundation for the entire machine and shall provide positive means for securing the column to the base. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed within the base. Precision finished guideways shall support and guide a saddle. Gibs or other means shall be provided to compensate for wear. Means for securing and leveling on a foundation shall be provided.

3.4.2 Column. The columns shall be a casting or weldment of box-type construction, internally ribbed and braced to minimize distortion, deflection and vibration. The overall assembled construction shall provide the required rigidity for accurate heavy metal removal cuts, and precision finish cuts under various machining conditions. Precision guideways shall provide full contact and support of the spindle head assembly throughout its travel range. Gibs or other means shall be provided to compensate for wear.

3.4.3 Spindle head. The spindle head for type I machines shall consist of the spindle motor and spindle drive mechanism, a vertically traversing saddle containing the spindle and quill assembly, saddle elevating mechanism, and a sliding shaft for transmission of power from the spindle motor to the saddle. The saddle shall be positioned vertically on guideways fitted to the spindle head. The spindle head shall traverse horizontally on ways of the fixed cross-rail. The spindle head for type II machines shall consist of the spindle motor, spindle drive mechanism and spindle and quill assembly. The spindle head shall be mounted on ways of the traveling cross-rail and traverse horizontally thereon. Vertical travel of the spindle head shall be provided by traverse of the cross-rail on ways of the columns. The spindle head or saddle shall be positioned by power actuated by the operator or control command. The spindle head shall be counterbalanced for ease of operations and conservation of energy. An automatically operated clamping device shall hold the head securely in the selected position.

3.4.4 Spindle and quill assembly. The spindle and quill assembly shall include the spindle, quill, and its related components. The spindle shall rotate and be supported in the quill on antifriction bearings. Spindle rotation shall be reversible for tapping operations. Unless otherwise specified (see 6.2.1), the spindle nose shall be the manufacturer’s standard quick tool change type, or when specified (see 6.2.1), collets, or spindle tool adapters shall be provided for holding the size and type taper and straight shank tools as specified by the procuring activity, The spindle assembly shall have power feed with provision for manual and tape selection of rapid and normal spindle positioning rate. Means shall be provided to prevent overtravel and damage to the feed mechanism. A spindle rotation

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locking device shall be provided and shall be interlocked so that the spindle start will not operate until the lock has been released. A direct leading indicator or other means shall be provided to clearly indicate the spindle speed and feed rate in use.

3.4.5 Spindle head and saddle. The spindle head and saddle shall be fabricated from a casting or weldment and be mounted on the cross-rail or guideways that accurate positioning is maintained throughout the range of travel . The saddle shall be equipped with protective covers for the positioning screw and guideways. Overtravel limit devices shall be provided at extremes of spindle head or saddle travel.

3.4.6 Worktable. The worktable shall be of cast construction, shall travel on the guideways on the foundation, and shall be accurately positioned within its range of travel by manual control devices and automatically by control command. The worktable shall be equipped with protective covers for the positioning screw and guideways. Automatic means shall be provided to prevent worktable movement after positioning. Overtravel protection devices shall be provided at extremes of worktable travel. A coolant trough shall be integral with the worktable with provisions for draining the coolant into the reservoir. T-slots running the full length of the work holding surface shall be provided; the number and spacing shall be the builders standard or as specified (see 6.2.1). When specified (see 6.2.1), the machine shall be equipped with a rotary table with means for manual or numerical control of rotary movement as required. Rotary table size and its corresponding characteristics shall be as specified by the procuring activity.

3.4.7 Spindle drive mechanism. The spindle drive mechanism shall be powered by an electric motor and shall have an arrangement to allow spindle speed change. The spindle drive shall be through belt, gears or direct motor drive. Drive to the slide units shall be by the manufacturer’s standard mechanism that together with the feedback system provided will be capable of achieving the position accuracy specified. Manual speed and feed change devices shall be located on the machine control, console or pendant. When operating in the automatic system, speed and feed changes shall be selected by control command. Means for adjusting drive motor belt tension shall be provided unless a non-tensioning, non-slipping type belt is provided.

3.4.8 Position measuring system. A precision, closed loop type measuring system shall be provided for precision positioning and measuring on each axis of movement. The measuring system for the size machine required shall be the manufacturer’s standard.

3.4.9 Lighting. A work light for illumination of the work area shall be provided. The light shall be shielded or protected type and shall provide for adjustability for position, or built-in type.

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3.4.10 Computer numerical control (CNC). The jig boring machine will be equipped with a fully automatic solid state soft-wired, integrated circuit type numerical control. The numerical control shall provide automatic control of machine functions, operating modes, spindle speeds and direction of rotation and simultaneous control of the machine's three linear axes in lnear, circular and helical interpolation and the rotary dividing table, if ordered (see 3.4.6). The control shall have the features and capabilities specified herein. Machine axes shall be identified in accordance with EIA-RS-267. Line voltage fluctuation of +10 percent from the normal operating voltage shall not adversely affect the operation of the CNC. Unless otherwise specified (see 6.2.1), the control shall be provided for connection of a 115-volt AC 60 hz, single phase power source. The control shall be capable of operating in an outside ambient temperature of 50-110 degrees F at 5-95 percent noncondensating relative humidity. The control shall be automatically shut down when the maximum operating temperature is reached. Program resolution shall be 0.0001 inch for each linear axis. The mechanical and electrical design of the control shall comply with EIA-RS-281 construction standards. The control shall be capable of accepting and processing input data in the form and format specified by the ordering activity. When EIA-RS-494 input data is specified, EIA Standards RS-227, RS-244, RS-274, and RS-358 do not apply to the numerical control.

3.4.10.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be 1-inch, eight-channel punched tape complying with EIA-RS-227.

3.4.10.1.1 Tape reader. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inch in diameter. The tape reader shall be housed in a dustproof enclosure mounted in the control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.10.2 Flexible diskette data input. When 32 Bit Binary Cutter Location (BCL) Exchange Input for numerically controlled machines is specified (see 6.2.1), the input media shall be an 8- inch flexible diskette recorded in accordance with 3.4.10.2.1. The input data format shall comply with EIA Standard RS-494. Manufacturers providing functions or features that are not

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currently covered in EIA-RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the manufacturer's features or functions. The control shall be equipped with an 8-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.10.2.1 Flexible diskette format. The 8-inch flexible disk for input of BCL data shall be a single density recording on one side using a soft-sectored format. The disk shall be divided into 77 distinct locations on the single-aide disk. Tracks shall be assigned numbers from 0 through 76. Track 0 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 0, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk that shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write bead slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26 addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 0 sector 1, designated as 0. The first part program shall be located at sector 1 E hexadecimal corresponding to track 1, sector 5. Where two-byte numbers are referenced, the high order byte shall be recorded first and the low order last. Track 1, sectors 1 through 4 shall contain the program directory of up to 16 part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end-of-file (EOF) records (8000 0000). Diskette addressing shall comply with the requirements of table I.

3.4.10.3 Program features.

3.4.10.3.1 Tape coding. When 1-inch punched tape is the specified input media, the control shall accept punched tape programmed in accordance with EIA-RS-274. The control shall be capable of recognizing and automatically accepting coding in accordance with EIA-RS-244 or EIA-RS-358.

3.4.10.3.2 Leading zero suppression. The control shall be capable of suppressing leading zeros. Not applicable to BCL control.

3.4.10.3.3 Selectable inch/metric input. The control system shall provide operator or program selectable Inch/metric input capability.

3.4.10.3.4 Absolute/incremental input. The control shall have switchable absolute/incremental input capability. The BCL control shall not be equipped with switchable absolute/incremental input.

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3.4.10.3.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute model shall be not less than plus or minus 99.9999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.10.3.6 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. Does not apply to BCL controls.

3.4.10.3.7 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274. Does not apply to BCL controls.

3.4.10.3.8 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Inches per minute (ipm)/millimeter per minute (mmpm).

3.4.10.4 Axis and tool management.

3.4.10.4.1 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system +0.0001" to 1.0000".
b. Metric system +0.001 mm to +25.400 mm.

3.4.10.4.2 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system +0.0002" to +1.0000".
b. Metric system +0.000 mm to +25.400 mm.

3.4.10.4.3 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.10.4.4 Reversal error compensation. Reversal error compensation circuitry shall be provided for a selectable dimension correction to be added automatically to the command dimension to each linear axis reversal to compensate for mechanical lost motion.

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3.4.10.4.5 Lead error compensation. Lead error compensation software shall be provided to permit compensating for the lead errors of each axis ball screw.

3.4.10.4.6 Auxiliary functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program command auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous function shall include program stop, optional program stop, end of program, spindle CW, spindle CCW, spindle off, coolant on-off, tool change, end of program (rewind), and program restart. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.4.10.5 Operator interface and controls.

3.4.10.5.1 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis (may be a LED display on the control unit).
d. Control and system error messages.

3.4.10.5.2 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, manual data input, sequence number search, and optional stop.

3.4.10.5.3 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.10.5.4 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.10.5.5 Block-by-block read. The control shall have block-by-block read function to provide for reading information one block at a time.

3.4.10.5.6 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequenced number.

3.4.10.5.7 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited pro-

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gram data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.10.5.8 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent.

3.4.10.5.9 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 and 120 percent of programmed speeds.

3.4.10.5.10 Mirror image. The control shall be capable of producing either right- or left-hand parts from the same tape data through the inversion of X and Y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.10.5.11 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus directions.

3.4.10.5.12 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.10.5.13 Tape tryout. The control shall have a tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.10.5.14 Operator control panel. The control unit shall have a control panel which provides, unless otherwise specified (see 6.2.1), at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and EOF Stop.

3.4.10.6 Miscellaneous control requirements.

3.4.10.6.1 Part program storage. Part program memory loadable from the data input media shall be provided. Unless otherwise specified (see 6.2.1), the part program memory shall be capable of storing not less than 100 feet of postprocessed part program tape.

3.4.10.6.2 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

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3.4.10.6.3 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.10.6.4 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program. The software shall be provided on 1-inch punched tape or 8-inch flexible disk as appropriate for the specified input media.

3.4.10.6.5 Maintenance diagnostic systems. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or in the form of 1-inch tape or 8-inch flexible disk, as required by the specified input media, to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.10.6.6 Peripheral equipment interface. The control shall be equipped with an interface complying with EIA Standard RS-232.

3.4.10.6.7 Canned cycles. Unless otherwise specified (see 6.2.1), the following canned cycles shall be provided:

a. Drilling
b. Boring
c. Tapping
d. Bolt hole circle

Such cycles shall be called by G codes and shall be initiated when X, Y, Z and R plane data is entered plus data for the F, E and K words.

3.4.10.6.8 Operator portable or pendant control. When specified (see 6.2.1), an operator's control of the pendant or portable box type shall be provided for the CNC with features as specified by the procuring activity.

3.4.10.6.9 Postprocessor. Unless otherwise specified (see 6.2.1), the CNC shall be furnished with postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified part program language (see 6.2.1) to the form and format required for input to the CNC. The postprocessor shall be written for use with the computer specified (see 6.2.1). Documentation sufficient to implement and use the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the jig boring machine.

3.5 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir

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shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.6 Pneumatic system. If compressed air is required for machine operation, the pneumatically controlled or operated components and the air distribution system shall comply with the requirements of JIC-P-1.

3.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.7.1 Motors. Motors shall be rated for a short duty cycle of not less than 60 minutes and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.7.2 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar aging devices is permitted only for voltage surge protection to other solid state components.

3.7.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.8 Size and capacity. Unless otherwise specified (see 6.2.1), the sizes and characteristics of the machine shall be not less than those specified in table II for the machine size specified.

3.9 Performance. The jig boring machine shall be capable of performing in accordance with the tests in 4.5.

3.9.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.10 Alignment tolerances. Each machine shall meet the alignment accuracies of table III.

3.11 Standard equipment. Unless otherwise specified one set of hand tools, normally furnished by the manufacturer, shall be furnished. Addi-

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tional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.12 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.13 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.14 Marking of instruction plates, indicating plates and name plates. All words on instruction plates, indicating plates and name plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wraparound tag, stencil, silk screen or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.14.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing internal
Type of lubricant
Viscosity

3.14.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine and contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.15 Hour meter. Unless otherwise specified (see 6.2.1) the machine shall be equipped with an hour meter. The hour meter shall be installed to

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display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to accumulate time.

3.16 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.17 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the rela-

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tive humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or the inspection shall be cause for rejection.

4.6 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.14.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment and positioning test. The machine shall be checked in accordance with the machine tool builder's standard test procedure to determine that the machine conforms to the accuracies of table III. Test instruments may be of the builder's choice, however, their accuracies shall be traceable to the Bureau of Standards.

4.7.3 Performance tests. The following test shall be performed to demonstrate the machine's capability of accurately positioning, boring and relocating. Under numerical control, drill, bore and counter bore three holes, 3/4-inch diameter by 1-inch deep, in a right angle pattern in a metal plate not less than 1-1/2-inch thick. Counter bored holes shall be 1-inch in diameter by 1/8-inch deep. The plate shall be of a material chosen by the supplier and both faces may be surface ground prior to the test. Predetermined dimensions between the bored holes shall be spaced to utilize as much of the normal working range of the machine as practical, allowing for setup equipment. After drilling, boring and counter boring operation is completed, the center to center dimensions between the holes shall be checked and shall be within ±0.00025 inch. Hole diameters shall be alike within 0.0003 TIR. Concentricity of bored and counter bored diameters shall be within 0.0002" TIR.

4.7.3.1 Positioning accuracy and repeatability. The positioning accuracy and repeatability for each axis of motion shall be within the tolerances of table III when evaluated in accordance with the NMTBA Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools,

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sections 1 and 4, Linear Definition. At least 10 random target points, spaced over the full length of the axis, shall be used for evaluating the accuracies. If the system uses a bi-directional programmed approach, each target point shall be evaluated from both directions. Should the axis length exceed the range of the instrument used, the zero position may be shifted to permit checking the full range of travel.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic Interference control testing shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Presentation, packing and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The numerically controlled jig boring machines covered by this specification are intended for use in any toolroom or production shop where precision boring, drilling, tapping and milling is required to meet work piece tolerances.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

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g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Electromagnetic interference control, if required; specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

j. Spindle nose type, collets, or tool adapters for straight shank and taper tools, and sizes required (see 3.4.4).

k. Table T-slots, specify if different (see 3.4.6).

l. Rotary table, if required, specify size and operating features (see 3.4.6).

m. CNC voltage and frequency of AC, if different (see 3.4.10).

n. One-inch punched tape data input, if required (see 3.4.10.1).

o. Tape reader spools, if different, specify (see 3.4.10.1.1).

p. 32 Bit Binary CL Exchange (BCL) Input data, if required (see 3.4.10.2).

q. Additional CNC features, if different, specify (see 3.4.10.3.8).

r. Auxiliary CNC functions, if different, specify (see 3.4.10.4.6).

s. Data display, if different, specify (see 3.4.10.5.1).

t. Operator control panel, if different, specify (see 3.4.10.5.14).

u. Part program memory, if different, specify (see 3.4.10.6.1).

v. Standby battery power, if different specify (see 3.4.10.6.2).

w. Canned cycles, if different specify (see 3.4.10.6.7).

x. Postprocessor documentation other than program(s), if required, specify (see 3.4.10.6.9).

y. Part programming language, specify (see 3.4.10.6.9).

z. Computer used for postprocessing, specify (see 3.4.10.6.9).

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aa. Size and capacity, if different (see 3.8).

bb. Additional performance requirements, specify (see 3.9.1).

cc. Equipment other than standard, if required, specify (see 3.11).

cc. Optional equipment, if required, specify (see 3.12).

ee. Repair parts, specify requirements (see 3.13).

ff. Lubrication chart or plate, if different, specify (see 3.14.1).

gg. Nameplate, if different, specify (see 3.14.2).

hh. Hour meter, delete if not required (see 3.15).

ii. First article inspection, if different specify (see 4.4).

jj. Quality conformance inspection, if different, specify (see 4.5).

kk. Level of preservation, packing, and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements (parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results) shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the

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exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.g.

6.5 Subject term (keyword listing).

Binary Cutter Location (BCL)
Boring machine
Computer numerical control
Fixed crossrail
Flexible diskette
Jig
Manual data input NC
Punched tape
Traveling crossrail

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TABLE I. Diskette addressing.

Sector address	Track Number	Sector Number	Byte	Use
0-25	0	All	All	Unused
26-29	1	1-4	0-1	File start sector address (2 byte binary)
			2-31	30 character ASCII program to be defined by user
			32-63	Same as 0-31 above for next part program
			64-95	Same as 32-63 above for next part program
			96-127	Same as 32-63 above for next part program
30-51	1	5-26	All	BCL Records for NC part programs
52-1923	2-73	All	All	BCL records for NC part programs
1924-2001	74-76	All	All	Unused

1/ This numbering corresponds to the 3740 physical fomat.

2/ Where 2-byte numbers are referenced, the high order byte is recorded first and low order last.

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TABLE II - Sizes and capacities.

Characteristics	Type I			
	Size 1	Size 2	Size 3	Size 4
Capacity:				
Table working surface inches width x length	30 x 24	30x34	43 x 33	63 x 48
Table travel inches longitudinal (x)	25	35	40	60
Spindle head travel, inches traverse on fixed or traveling cross rail (y)	18	23	28	40
Table top to spindle nose range in inches	3/8 - 26-1/4	3/8 - 33-1/2	0 - 40	2-3/8 - 49-3/4
Spindle head travel on columns, inches vertical (w)	19-1/2	25	30-3/4	35-3/8
Spindle head travel on spindle carrier, inches vertical (w)	—	—	—	—
Spindle quill travel, inches vertical (z)	6-1/4	7-3/4	12	12
Distance between uprights	29	37	43	63
Speeds and feeds:				
Spindle speeds number	infinite	infinite	infinite	infinite
Spindle speed range rpm	10-800 or 800-2500	10-80 or 800-2500	30-280 or 280-830 or 830-2500	30-280 or 280-830 or 830-2500
Spindle feeds, number and range IPR or IPM	infinite 0.0004-0.0150 IPR	infinite 0.0004-0.0150 IPR	infinite 0.0004-0.0150 IPR	infinite 0.0004-0.0150 IPR
Head rapid traverse, ipm	230	230	230	230
Worktable and saddle, worktable XY	230	230	230	230
saddle W	31	31	230	230
Worktable and saddle mill rate, range, ipm	1 to 15	1 to 15	1 to 15	1 to 15
Motors, horsepower:				
Spindle drive	6	6	8	8

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above tables. When the largest size is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE III - Alignment and positioning accuracies.

Requirements	Permissible tolerance
Concentricity of spindle nose internal bore, 3/8 inch for nose	0.00005" TIR
Concentricity of spindle bore (runout) proving bar in spindle nose bore	
2 inches from nose	0.00013" TIR
6 inches from nose	0.00025" TIR
Worktable trueness of travel	0.0002" overall
Spindle axis square with worktable surfaces (tramming method) largest practicable	0.00032", 8" radius or less
diameter or tramming circle	0.00053" over 8" radius
Quill motion square to worktable surface (Head locked before each reading)	
Quill movement alone	0.00010"/10"
Head movement alone	0.00010"/10"
Head and quill combined movement	0.00020" overall
Front of worktable parallel with longitudinal motion	0.00015" TIR
T-slots parallel to each other	0.001"/ft.
Longitudinal motion square to transverse motion	0.0001"/ft. 0.0003" overall
Worktable flatness (flat to concave only) Crosswise 0.0001"/ft.	
Over full width	0.0002"
Lengthwise	0.0002"/ft.
Over full length	0.0002"
Worktable and spindle positioning accuracy Positioning accuracy of X and Y axes, bi-directional approack, over entire travel of axis at Sigma 3 value of the frequency distribution	0.00012"
Worktable and spindle positioning repeatability repeatability, bi-directional approach over entire travel of X and Y axes	10 cycles ±0.00006"
Depth axis repeatability (Z)	±0.0002"
Vertical rotary table "A" axis option	
Positioning accuracy	2 arc seconds
Repeatability	1 arc second

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AR
Air Force - 84
Navy - YD

User Activities:
Army - AL
Air Force - 84

Preparing Activity:
DLA-IP

Project 3411-0068

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80272	Boring Machines, Jig Vertical, Double Column, Bridge Type, Fixed or Traveling Cross-rail, Numerically Controlled

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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